The Continuous-Time Fourier Transform

1 From Periodic to Aperiodic Signal

Consider an aperiodic signal $x(t)$ that is nonzero only over $[0, T_1]$. There is a corresponding signal $\tilde{x}(t)$ with period T that is the same as $x(t)$ on $[0, T_1]$ and $T \geqslant T_1$.

$$x(t) = \begin{cases} x(t) & \text{for} \quad t \in [0, T_1] \\ 0 & \text{otherwise} \end{cases} \qquad \tilde{x}(t + kT) = x(t) \quad \text{for} \quad k \in \mathbb{Z} \quad \text{and} \quad t \in [0, T] \qquad T_1 \leqslant T$$

Intuitively, we have

$$x(t) = \lim_{T \to \infty} \tilde{x}(t)$$

The Fourier coefficients for the periodic signal $\tilde{x}(t)$ are

$$\tilde{a}_k = \frac{1}{T} \int_0^T \tilde{x}(t) \mathrm{e}^{-\mathrm{j}k\omega_0 t} \, \mathrm{d}t = \frac{1}{T} \int_0^T x(t) \mathrm{e}^{-\mathrm{j}k\omega_0 t} \, \mathrm{d}t = \frac{1}{T} \int_{-\infty}^{\infty} x(t) \mathrm{e}^{-\mathrm{j}k\omega_0 t} \, \mathrm{d}t$$

We define

$$X(\mathrm{j}\omega) := \int_{-\infty}^{\infty} x(t) \mathrm{e}^{-\mathrm{j}\omega t} \, \mathrm{d}t \qquad X(\mathrm{j}k\omega_0) = \int_{-\infty}^{\infty} x(t) \mathrm{e}^{-\mathrm{j}k\omega_0 t} \, \mathrm{d}t = T\tilde{a}_k$$

Then

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} \tilde{a}_k \mathrm{e}^{\mathrm{j}k\omega_0 t} = \sum_{k=-\infty}^{\infty} \frac{1}{T} X(\mathrm{j}k\omega_0) \mathrm{e}^{\mathrm{j}k\omega_0 t} = \sum_{k=-\infty}^{\infty} \frac{\omega_0}{2\pi} X(\mathrm{j}k\omega_0) \mathrm{e}^{\mathrm{j}k\omega_0 t}$$

$$\begin{aligned} x(t) &= \lim_{T \to \infty} \tilde{x}(t) \\ &= \lim_{\omega_0 \to 0^+} \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(\mathrm{j}k\omega_0) \mathrm{e}^{\mathrm{j}k\omega_0 t} \omega_0 \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\mathrm{j}\omega) \mathrm{e}^{\mathrm{j}\omega t} \, \mathrm{d}\omega \end{aligned}$$

Definition 1.1 (The Fourier Transform Pair of a CT Aperiodic Signal)**.**

If an aperiodic signal $x(t)$ has a Fourier transform, then

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\mathrm{j}\omega) \mathrm{e}^{\mathrm{j}\omega t} \, \mathrm{d}\omega \qquad \text{(The Synthesis Equation)}$$

$$X(\mathrm{j}\omega) = \int_{-\infty}^{\infty} x(t) \mathrm{e}^{-\mathrm{j}\omega t} \, \mathrm{d}t \qquad \text{(The Analysis Equation)}$$

where the function $X(\mathrm{j}\omega)$ is called the *Fourier transform* or *Fourier integral* of $x(t)$, and the *synthesis equation* is called the *inverse Fourier transform*.

The Fourier coefficients of the periodic signal $\tilde{x}(t)$ are proportional to spaced samples of the Fourier transform of the finite-duration signal $x(t)$:

$$\tilde{a}_k = \frac{1}{T} X(\mathrm{j}\omega) \Big|_{\omega = k\omega_0}$$

For aperiodic signals, $X(\mathrm{j}\omega) \dfrac{\mathrm{d}\omega}{2\pi}$ can be seen as "amplitude" of the complex exponential compoents. Therefore, the transform $X(\mathrm{j}\omega)$ is referred to as the *spectrum* of $x(t)$.

2 Fourier Transform for Periodic Signals

Theorem 2.1.

For a periodic signal $x(t)$ whose FS coefficients are $\{a_k\}$, the Fourier transform $X(\mathrm{j}\omega)$ is given by

$$\begin{aligned} X(\mathrm{j}\omega) &= \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0) \\ &= \begin{cases} 2\pi a_k \delta(0) & \text{for } \frac{\omega}{\omega_0} = k \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

***Proof*.**

$$\begin{aligned} X(\mathrm{j}\omega) \quad \xrightarrow{\mathcal{F}^{-1}} \quad \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\mathrm{j}\omega)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega \\ &= \sum_{k=-\infty}^{\infty} a_k \int_{-\infty}^{\infty} \delta(\omega - k\omega_0)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega \\ &= \sum_{k=-\infty}^{\infty} a_k \mathrm{e}^{\mathrm{j}k\omega_0 t} \\ &= x(t) \end{aligned}$$

□

3 Properties of CT Fourier Transform

3.1

Suppose we have:

$$\begin{cases} x(t) \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} X(\mathrm{j}\omega) \\ y(t) \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} Y(\mathrm{j}\omega) \end{cases}$$

Then we have the following results:

Linearity

$$a \cdot x(t) + b \cdot y(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad a \cdot X(\mathrm{j}\omega) + b \cdot Y(\mathrm{j}\omega)$$

Time Shifting

$$x(t - t_0) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad \mathrm{e}^{-\mathrm{j}\omega t_0} X(\mathrm{j}\omega)$$

Frequency Shifting

$$\mathrm{e}^{\mathrm{j}\omega_0 t} x(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad X(\mathrm{j}(\omega - \omega_0))$$

Time Reversal

$$x(-t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad X(-\mathrm{j}\omega)$$

Scaling

$$x(at) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad \frac{1}{|a|} X(\frac{\mathrm{j}\omega}{a})$$

Conjugation

$$x^*(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad X^*(-\mathrm{j}\omega)$$

Convolution

$$x(t) * y(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad X(\mathrm{j}\omega)Y(\mathrm{j}\omega)$$

Multiplication

$$x(t)y(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad \frac{1}{2\pi}X(\mathrm{j}\omega) * Y(\mathrm{j}\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\mathrm{j}\theta)Y(\mathrm{j}\omega - \mathrm{j}\theta)\,\mathrm{d}\theta$$

Differentiation in Time

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad \mathrm{j}\omega X(\mathrm{j}\omega)$$

Integration in Time

$$\int_{-\infty}^{t} x(\tau)\,\mathrm{d}\tau = x(t) * u(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad \frac{1}{\mathrm{j}\omega}X(\mathrm{j}\omega) + \pi X(0)\delta(\omega)$$

Differentiation in Frequency

$$t \cdot x(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad \mathrm{j}\frac{\mathrm{d}}{\mathrm{d}\omega}X(\mathrm{j}\omega)$$

Even-Odd Decomposition for Real Signals

$$x(t) \text{ real:} \quad \mathcal{E}v\{x(t)\} \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad \mathcal{R}e\{X(\mathrm{j}\omega)\}$$

$$x(t) \text{ real:} \quad \mathcal{O}d\{x(t)\} \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad \mathrm{j}\,\mathcal{I}m\{X(\mathrm{j}\omega)\}$$

Parseval's Relation for Aperiodic Signals

$$\int_{-\infty}^{\infty} |x(t)|^2\,\mathrm{d}t \quad = \quad \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\mathrm{j}\omega)|^2\,\mathrm{d}\omega$$

3.2 Symmetry

- $x(t)$ real $\quad \Longleftrightarrow \quad X(\mathrm{j}\omega) = X^*(-\mathrm{j}\omega)$
- $x(t)$ purely imaginary $\quad \Longleftrightarrow \quad X(\mathrm{j}\omega) = -X^*(-\mathrm{j}\omega)$

Conjugate Symmetry For real $x(t)$, its FT $X(\mathrm{j}\omega)$ satisfies

- $X(\mathrm{j}\omega) = X^*(-\mathrm{j}\omega)$
- $\mathcal{R}e\{X(\mathrm{j}\omega)\} = \mathcal{R}e\{X(-\mathrm{j}\omega)\}$
- $\mathcal{I}m\{X(\mathrm{j}\omega)\} = -\mathcal{I}m\{X(-\mathrm{j}\omega)\}$
- $|X(\mathrm{j}\omega)| = |X(-\mathrm{j}\omega)|$
- $\arg X(\mathrm{j}\omega) = -\arg X(-\mathrm{j}\omega)$

Parity

- $x(t)$ even $\quad \Longleftrightarrow \quad X(\mathrm{j}\omega)$ even
- $x(t)$ odd $\quad \Longleftrightarrow \quad X(\mathrm{j}\omega)$ odd
- $x(t)$ real & even $\quad \Longrightarrow \quad X(\mathrm{j}\omega)$ real & even
- $x(t)$ real & odd $\quad \Longrightarrow \quad X(\mathrm{j}\omega)$ purely imaginary & odd

3.3 Duality

$$f(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad g(\omega) \qquad \Longleftrightarrow \qquad g(t) \quad \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftharpoons}} \quad 2\pi \cdot f(-\omega)$$

4 Basic Fourier Transform Pairs

$$\mathrm{sinc}(\theta) := \frac{\sin(\pi\theta)}{\pi\theta} \qquad \mathrm{rect}(t) := u(t+\frac{1}{2}) - u(t-\frac{1}{2}) = \begin{cases} 1 & \text{for} \quad |t| < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

Signal	Fourier Transform	Fourier Series Coefficients
$\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	$\begin{cases} a_1 = 1 \\ a_k = 0 \quad \text{otherwise} \end{cases}$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$\begin{cases} a_1 = a_{-1} = \frac{1}{2} \\ a_k = 0 \quad \text{otherwise} \end{cases}$
$\sin(\omega_0 t)$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$\begin{cases} a_1 = a_{-1} = \frac{1}{2j} \\ a_k = 0 \quad \text{otherwise} \end{cases}$
Periodic Square Wave $\begin{cases} 1 & \lvert t\rvert < T_1 \\ 0 & T_1 < \lvert t\rvert \leqslant T/2 \end{cases}$ with period of T	$\sum_{k=-\infty}^{\infty} \frac{2\sin(k\omega_0 T_1)}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \mathrm{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin(k\omega_0 T_1)}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$
$\mathrm{rect}\left(\frac{t}{2T_1}\right) = \begin{cases} 1 & \lvert t\rvert < T_1 \\ 0 & T_1 < \lvert t\rvert \end{cases}$	$\frac{2\sin\omega T_1}{\omega} = 2T_1\,\mathrm{sinc}\left(\frac{T_1\omega}{\pi}\right)$	
$\frac{\sin Wt}{\pi t}$	$\begin{cases} 1 & \lvert\omega\rvert < W \\ 0 & W < \lvert\omega\rvert \end{cases}$	
$\mathrm{rect}(t)$	$\frac{\sin(\omega/2)}{\omega/2} = \mathrm{sinc}\left(\frac{\omega}{2\pi}\right)$	
$\mathrm{sinc}\left(\frac{\omega}{2\pi}\right)$	$2\pi\,\mathrm{rect}(t)$	
1	$2\pi\delta(\omega)$	$\begin{cases} a_0 = 1 \\ a_k = 0 \quad \text{otherwise} \end{cases}$ (For any choice of $\omega_0 > 0$)
$\delta(t)$	1	
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$	
$\delta(t - t_0)$	$e^{-j\omega t_0}$	
$e^{-at}u(t)$ with $\mathcal{Re}\{a\} > 0$	$\frac{1}{a + j\omega}$	
$te^{-at}u(t)$ with $\mathcal{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	
$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t)$ with $\mathcal{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	

5 Systems Characterized By LCCDEs

An LTI system with initial rest characterized by the LCCDE

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

can be represented in the frequency domain as

$$H(\mathrm{j}\omega) = \frac{Y(\mathrm{j}\omega)}{X(\mathrm{j}\omega)} = \frac{\displaystyle\sum_{k=0}^{M} b_k (\mathrm{j}\omega)^k}{\displaystyle\sum_{k=0}^{N} a_k (\mathrm{j}\omega)^k}$$